E-Portfolio Reflection

In units 1, 2, 3, 4, 9, 10 and 12. I did lecture casts on different things relating to the assignments. These consisted of data sources and methods of statistical measurement, data structures, data management in R, probabilities, creating cross tabulations and performing chi-square analysis, correlation and regression and bayesian data analysis. These lecture casts gave me lots of important information, lots of which I did not know before doing these lecture casts. This information was useless throughout this module. It felt a little bit overwhelming at first but once I started to understand it more and more it all started to make sense. This made me feel good because I was finally understanding what I had to do in the assignments.

In unit 5 I had a discussion topic with other students about descriptive statistics and visualisation. This was to allow us to have an open discussion between all students and give our own opinions on this topic. The purpose of this was to critically evaluate the use of statistical analysis and the numeric interpretation of results as aids in the decision making process and critically appraise and present results of a statistical analysis to a diverse audience. I found this task a bit challenging to begin with but as I managed to understand it more it became easier to accomplish and know what was expected of me.

Unit 6 was a statistics quiz with 5 questions all relating to R and data, I found these a little tricky, some were simple but some took me a while to understand, this made me feel a little stressed but once I got to grips with it, it was useful to know this information for future reference when working with R.

Unit 7 consisted of a mathematics test which was 25% towards the module mark. The test consisted of different mathematics questions revolving data and to prepare for this the lectures were very helpful. I found this quiz very easy to do and was a little nervous towards doing it but I felt well prepared.

In unit 8 I did a scenario based exercise on 95% confidence interval where I had to work out the mean efficiency of three vendors and show which one has the higher significance of improvement

in employee efficiency statistically at the 95%| level. This was a struggle to begin with but after some time and research I managed to figure out what I needed to do, this was quite interesting to learn and know, the learning outcome for this was to be able to apply mathematical and statistical methods in these fields to help decision making, also to critically evaluate the use of statistical analysis and the numeric interpretation of results to aid in decision making.

Unit 9 discussion 2 about misinterpretation of statistical information involved having a discussion with other students and answering two questions, under what situations should confidence intervals not be reported? and why is p value not always enough to report? The purpose of this discussion was to understand why we report certain values and what they mean and what happens when misinterpretations occur. I found this information very useful for understanding why I was interpreting the values I was during this module.

In units 1,2,3,4,5,7, 9, 10, 11 I did data activities, these were to prepare me for the unit 12 assignment. Using a crime survey for England and Wales 2013-2014 I had to perform different tasks in each unit in R to prepare me for the assignment. I did struggle a lot with some of these tasks but once I understood them doing the assignment was much easier.

e survey for England and Wales 2013-2014 and I had to create a summary statistic using the antisocial variable, I found this challenging at first as I had never worked with R and I was still learning it, I did some research and gathered the information I needed to be able to perform this task. After doing this task I felt more confident moving forward to the second data activity. The learning outcome of this task is to have a systematic understanding of the key mathematical and statistical concepts and techniques which underpin mechanisms in data science and AI.

In my Unit 12 reflection assignment this allowed me to gain an understanding of what I struggled with throughout the module and what I needed to work on, I now know what I need to work on and how comfortable I am with all the other tasks in R.

In the Unit 12 statistical analysis presentation I struggled with some of the questions, especially the statistical analysis and it took me a while to know what to do, I had to do some research and

watch some videos, and after some trial and error I managed to do it, this caused me a lot of stress and frustration but I managed to get there in the end.

I think after finishing this module I am much more confident in my ability to perform statistical data analysis in R, I have gained much more knowledge around it and understand the code more. I faced a lot of challenges throughout this module with struggles to understand certain tasks and some code not working and I had to figure out what I did wrong, sometimes I would have to scrap it and start fresh to figure out what I did wrong. After learning from all of these mistakes I am much more happy with working with R in the future. It took a lot of understanding on how to interpret statistical findings using the p value and confidence interval and I had to do some background research and get guidance but after this module I believe I am confident to do this again and I am much more happy with my outcome. While using R to create tables and plots I found this the easiest to understand, after my trouble understanding the variables I could use I found this relatively easy.